Week 1: Meet Your Robot

You will write MATLAB code that drives a small robot on the screen. This week the robot draws shapes; next week it uses **sensors** to find its way through tunnels and mazes. No programming experience is needed: every step is explained, and the computer checks your work. Each time you finish a task, MATLAB gives you a **mission code**. Type it in the Week 1 answer sheet (a Word file).

What you hand in

Open `Week1_Answer_Sheet.docx`. Type your name, every **mission code** exactly as MATLAB prints it, and your answers to the exit questions. Save it and upload it to the course page before you leave.

Setup (first 10 minutes)

Do these steps in order. After each step, check that you see what the box on the right describes.

	Do this	You should see
1	Download `TurtleRobot.zip` from the course page. *Can't?* Go to `https://github.com/mojafari/ENGR1701-Turtle-Robot`, click the green **Code** button, then **Download ZIP**.	The file in your **Downloads** folder.
2	**Unzip it.** Windows: right-click the file → **Extract All** → **Extract**. Mac: double-click it.	A normal folder called `TurtleRobot` (not a zip). From GitHub, `TurtleRobot` is inside a folder ending in `-main`.
3	Start **MATLAB R2025a** (Start menu or Dock). The first start can take 1–3 minutes. If it asks you to sign in, use your university email. If it asks about a license, raise your hand.	The MATLAB window with a **Command Window** (the area where it says »).
4	Find the panel that lists files (called **Current Folder** or **Files**). Use its folder/browse button to open your `TurtleRobot` folder.	The files `Turtle.m`, `checkMission.m`, `Week1_Meet_Your_Robot.m`, ...
5	Double-click `Week1_Meet_Your_Robot.m`.	The file opens in the **Editor** (green and black text).
6	Click inside the section **STEP 0** and press `Ctrl`+`Enter` (Mac: `Cmd`+`Enter`).	"SETUP OK!" in the Command Window and a robot that draws a green triangle.

Stuck? Common setup problems

- **"Unrecognized function or variable 'turtleSetupCheck'"** or a pop-up that says **Change Folder**: MATLAB is looking in the wrong folder. Click **Change Folder**, or repeat step 4.
- **Nothing happens when I double-click the zip**, or files look "inside" the zip: you skipped step 2. Extract first.
- **I can't see the robot**: the robot window is a tab called **Turtle Robot** inside MATLAB (or on the taskbar). Click it.

How to work in the files

- The files are split into **sections**. A section starts with a line that begins with `%%`. Click inside a section (it turns light yellow) and press `Ctrl`+`Enter`, or click **Run Section** on the **Editor** tab. Only that section runs.
- **Do not** press the big green **Run** button (`F5`). It runs the whole file and stops at the first unfinished task.
- Green lines starting with `%` are **comments**: instructions for you. MATLAB ignores them.
- Do the sections **in order**. Type your code on the empty lines under "your code here."
- A robot stuck in a loop that runs forever? Click in the Command Window and press `Ctrl`+`C`.

Red text means an error. That's normal!

Every programmer sees errors all day. MATLAB tells you *which line* and *what* went wrong. Read the message, fix that line, and run the section again.

If you see...	It usually means...
`Unrecognized method, property, or field 'foward' for class 'Turtle'`	A command is misspelled, or has the wrong capital letters. MATLAB is case sensitive: `left` works, `Left` does not.
`Unrecognized function or variable 't'`	You skipped `t = Turtle();` or ran a section out of order.
`Unrecognized function or variable 'fifty'`	You typed a word where MATLAB needs a number or a variable you created earlier.
`Invalid expression` / `Parenthesis is not balanced`	A missing or extra `( )`. Every `(` needs a `)`.
`CRASH! The robot hit a wall...`	Your robot drove into a wall. Count the cells again or use a sensor.
`Stopped: ...Is your loop running forever?`	Your `while` loop never ends. Check that the robot moves inside the loop.
`TRY AGAIN: ...`	The code ran, but the result isn't right yet. Read the hint after it.

Robot command cheat sheet

Your robot is a variable called `t`. Every command starts with `t.` (t, then a dot).

Command	What it does
`t = Turtle();`	Make a new robot on an empty canvas (Week 1).
`t = Turtle("maze");`	Make a new robot in a level: `"hallway"`, `"corner"`, `"stairs"`, `"tunnel"`, `"mystery"`, `"maze"`.
`t.forward(100)` / `t.backward(50)`	Drive forward / backward. In mazes the number is cells.
`t.left(90)` / `t.right(45)`	Turn left / right by an angle in degrees. `t.left()` turns 90.
`t.penUp()` / `t.penDown()`	Stop / start drawing while moving.
`t.setColor("red")`	Pen color: red, blue, green, orange, purple, black, ...
`t.setWidth(4)` / `t.setSpeed(10)`	Thicker pen / faster robot (1 = slow, 10 = fast).
`t.setRobotColor("blue")`	Change the color of the robot itself (not the pen).
`t.wallAhead()`, `t.wallLeft()`, `t.wallRight()`	Sensors: `true` if there is a wall there.
`t.distanceAhead()`	Sensor: number of free cells in front of the robot.
`t.atGoal()`	Sensor: `true` if the robot is on the goal.
`t.report()`	Print how many commands the robot used.
`checkMission(t, "square")`	Check your work. A PASS gives you a mission code.

Three ideas you will use

Repeat N times

```
for k = 1:4
    t.forward(100)
    t.left(90)
end
```

Repeat while true

```
while ~t.wallAhead()
    t.forward(1)
end
```

~ means **NOT**.

Decide

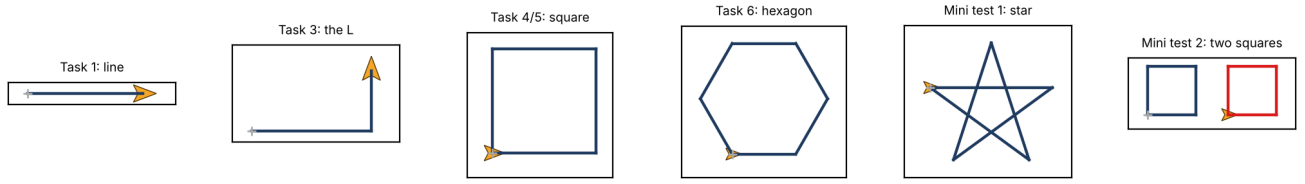
```
if t.wallLeft()
    t.right()
else
    t.left()
end
```

Week 1 – Meet your robot

File: `Week1_Meet_Your_Robot.m`. Work through the sections from top to bottom.

Section	What you learn
Step 0	Check that everything works.
Part A	MATLAB as a calculator; **variables** like `side = 100`; what the semicolon `;` does.
Task 1	Your first robot and your first command.
Task 2	Turning. Change the numbers and see what happens (no check).
Task 3	**Debugging**: fix three bugs by reading the error messages.
Task 4	Draw a square, one command at a time.
Task 5	Draw a square with a **for loop** and a variable.
Task 6	Any polygon: use the formula `angle = 360 / n`.
Mini test 1–2	Write the code yourself: a star, then two squares in two colors.
Bonus	Be creative: a spiral, a house, your initials.

What the shapes should look like (the + is where the robot starts):



Tips for Week 1

- Each task starts with `t = Turtle();`, which clears the screen. If a check says you drew too many lines, run the *whole* section again.
- Turning **left** is counterclockwise, turning **right** is clockwise. Both work for the shapes.
- Star hint: the robot turns 144 degrees at each point. Why 144? Five turns of 144 add up to 720, two full circles!